

Hospital Readmission Analytics

Diabetes Patient Readmission Analysis

Portfolio Analytics Project | Python • SQL • Power BI

Project Focus: Identifying factors associated with 30-day hospital readmission among patients with diabetes.

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Dataset: 101,766 hospital encounters | 71,518 unique patients

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Primary Outcome: 30-day hospital readmission

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Overall Readmission Rate: 11.16%

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1. Executive Summary

Hospital readmissions are an important healthcare quality and utilization indicator. This project analyzes a historical dataset of hospital encounters involving patients with diabetes to identify factors associated with 30-day hospital readmission.

The analysis follows an end-to-end healthcare analytics workflow:

Python → Data Quality & EDA → SQL → Power BI → Business Insights

The dataset contains **101,766 hospital encounters involving 71,518 unique patients**.

The overall encounter-level 30-day readmission rate is:

11.16%

At the patient level, **8,834 patients** experienced at least one 30-day readmission, corresponding to a patient-level readmission rate of:

12.35%

The strongest observed association was with prior inpatient utilization. Encounters involving patients with **4+ prior inpatient visits** had a 30-day readmission rate of **30.70%**, compared with the overall benchmark of **11.16%**.

Prior emergency utilization showed a similar pattern. Encounters involving patients with **3+ prior emergency visits** had a readmission rate of **24.94%**.

These findings indicate that previous healthcare utilization and patient complexity are important characteristics associated with subsequent readmission in this dataset.

However, these are **observational associations rather than causal relationships**. The dataset is historical and represents U.S. healthcare encounters from approximately 1999–2008, so the findings should not be interpreted as directly representative of current hospitals or patient populations in India.

Executive KPI Snapshot

KPI	Result
Total Encounters	101,766
Unique Patients	71,518
30-Day Readmissions	11,357
Overall 30-Day Readmission Rate	11.16%
Patients With ≥ 1 30-Day Readmission	8,834
Patient-Level Readmission Rate	12.35%
Patients With 2+ 30-Day Readmissions	1,539

Key Findings at a Glance

- **Prior inpatient utilization:** Readmission increased from **8.44%** with no prior inpatient visits to **30.70%** with 4+ prior inpatient visits.
- **Prior emergency utilization:** Readmission increased from **10.47%** with no prior emergency visits to **24.94%** with 3+ prior emergency visits.
- **Repeat readmission:** **1,539 patients** experienced 2 or more 30-day readmissions.
- **Length of stay:** Readmission generally increased across longer length-of-stay groups, reaching **13.50%** for 7–9 days.
- **Medication burden:** Readmission increased from **7.48%** in the 1–5 medication group to **12.87%** in the 21–30 group.
- **Diagnosis burden:** Readmission increased from **7.71%** for 1–4 diagnoses to **14.78%** for 10+ diagnoses, although the 10+ group contains only 115 encounters.

2. Business Problem

Business Question

What factors are associated with 30-day hospital readmission among patients with diabetes, and what insights can healthcare organizations derive from the data?

Business Objectives

The analysis aims to:

1. Measure the overall 30-day hospital readmission rate.
2. Measure readmission at both encounter and patient levels.
3. Identify patient and encounter characteristics associated with higher readmission rates.
4. Examine the relationship between previous healthcare utilization and readmission.
5. Analyze demographic, treatment, admission, discharge, and utilization characteristics.
6. Build an interactive Power BI dashboard for decision-oriented reporting.

7. Translate analytical findings into actionable healthcare insights while recognizing the limitations of observational data.

Tools & Skills Demonstrated

Area	Tools / Skills
Data Analysis	Python, pandas, NumPy
Data Quality	Missing-value assessment, validation, data profiling
Exploratory Analysis	Grouping, segmentation, rate analysis, feature engineering
Statistical Thinking	Benchmark comparison, subgroup analysis, association analysis
SQL	SQLite, aggregation, filtering, grouping, analytical queries
Business Intelligence	Microsoft Power BI
Dashboarding	KPI cards, benchmark lines, interactive analytical visuals
Predictive Analytics	Logistic regression, ROC-AUC, Average Precision, Brier Score
Documentation	Markdown, Visual Studio Code
Healthcare Analytics	Readmission analysis, utilization analysis, patient-level vs encounter-level metrics

3. Dataset Overview

The project uses the **Diabetes 130-US hospitals for years 1999–2008** dataset.

The dataset contains hospital encounter-level information for patients with diabetes, including:

- Patient identifiers
- Encounter identifiers
- Demographics
- Admission information
- Discharge information
- Previous outpatient visits
- Previous emergency visits
- Previous inpatient visits
- Length of hospital stay
- Number of laboratory procedures
- Number of procedures
- Number of medications
- Number of diagnoses
- Medication information
- Insulin treatment
- Medication changes

- Diabetes medication status
- Readmission status

Dataset Size

Metric	Value
Total encounters	101,766
Unique patients	71,518
30-day readmission encounters	11,357
Patients with ≥ 1 30-day readmission	8,834
Encounter-level readmission rate	11.16%
Patient-level readmission rate	12.35%

Dataset Source & Citation

The analysis uses the **Diabetes 130-US hospitals for years 1999–2008** dataset, containing hospital encounters for patients with diabetes.

The dataset was originally made available through the **UCI Machine Learning Repository** and has been widely used for research and educational analysis of hospital readmission.

Dataset: Diabetes 130-US hospitals for years 1999–2008

Observation unit: Hospital encounter

Primary outcome: Readmission within 30 days

Time period: 1999–2008

Healthcare setting: U.S. hospitals

The dataset is used in this project for portfolio and analytical-learning purposes. Results should not be interpreted as representing current hospital readmission patterns, current U.S. healthcare populations, or Indian healthcare populations.

Citation

Strack, B., De Shazo, J. R., Gennaro, M., et al. (2014). *Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records*. **BioMed Research International**, 2014.

The dataset is available through the UCI Machine Learning Repository under the title **Diabetes 130-US hospitals for years 1999–2008**.

4. Data Quality Assessment

Data quality assessment was performed in Python before analytical modeling and dashboard development.

The original dataset contains several missing-value representations and high-missingness variables.

The placeholder value ? was treated as missing rather than as a valid category.

Important Missing or Unknown Fields

Field	Missing / Unknown Records
Weight	98,569
Max glucose serum	96,420
A1C result	84,748
Medical specialty	49,949
Payer code	40,256
Race	2,273
Diagnosis 3	1,423
Diagnosis 2	358
Diagnosis 1	21

Weight, maximum glucose serum, and A1C result contained very high levels of missingness and were therefore excluded from the primary analysis.

The raw dataset was preserved separately so that the original source data remained unchanged.

Data Quality Snapshot

The source dataset contains several variables with substantial missing or placeholder values. Missingness was assessed before analytical preparation.

Variable	Missing / Placeholder Records	Approx. Share
weight	98,569	96.86%
max_glu_serum	96,420	94.75%
A1Cresult	84,748	83.28%
medical_specialty	49,949	49.08%
payer_code	40,256	39.56%
race	2,273	2.23%
diag_3	1,423	1.40%
diag_2	358	0.35%
diag_1	21	0.02%

Variables with extremely high missingness, particularly weight, max_glu_serum, and A1Cresult, were excluded from the primary analysis rather than attempting to impute large amounts of unavailable information.

The original dataset was preserved unchanged, while analytical transformations were performed on derived datasets.

5. Data Preparation

The analytical dataset was prepared using Python.

The main preparation steps included:

- Loading the original CSV dataset.
- Creating a copy for analysis.
- Converting ? placeholders to missing values.

- Removing variables with extremely high missingness from the primary analysis.
- Creating a binary 30-day readmission indicator.
- Creating utilization groups.
- Creating length-of-stay groups.
- Creating medication-burden groups.
- Creating diagnosis-burden groups.
- Creating laboratory-procedure groups.
- Creating admission and discharge descriptive categories.
- Creating diagnosis categories for exploratory analysis.

30-Day Readmission Definition

The target variable was defined as:

- 1 = patient was readmitted within 30 days (<30)
- 0 = patient was not readmitted within 30 days

The original readmitted field was retained for reference.

Methodology

The analysis was conducted in five main stages.

Stage 1 — Data Validation

Python was used to validate the structure and integrity of the source dataset.

Checks included:

- Dataset dimensions
- Column structure
- Unique patients
- Unique encounters
- Duplicate encounter IDs
- Readmission distribution
- Demographic distributions
- Missing and placeholder values

Stage 2 — Data Quality & Preparation

The dataset was prepared for analysis by:

- Converting ? placeholders to missing values.
- Assessing missingness across variables.
- Excluding variables with extremely high missingness from the primary analysis.
- Creating the binary readmitted_30d outcome.
- Creating utilization and burden groups.
- Creating descriptive admission, discharge, and diagnosis categories.

The original source dataset was preserved unchanged.

Stage 3 — Exploratory Data Analysis

Readmission rates were calculated across:

- Prior inpatient utilization
- Prior emergency utilization
- Prior outpatient utilization
- Age
- Gender
- Race
- Length of stay
- Medication burden
- Diagnosis burden
- Laboratory procedure burden
- Insulin treatment
- Medication changes
- Diabetes medication status
- Admission characteristics
- Discharge characteristics

Subgroup rates were compared with the overall **11.16% encounter-level benchmark**.

Stage 4 — SQL Validation

SQLite was used as an independent analytical layer.

Key findings from Python were reproduced using SQL, including:

- Overall readmission rate
- Patient-level readmission rate
- Prior inpatient utilization
- Prior emergency utilization
- Repeat readmission burden
- Benchmark comparisons

This provided an additional validation step before dashboard development.

Stage 5 — Power BI Reporting

Power BI was used to create an interactive reporting layer containing:

- Executive KPI cards
- Readmission-rate comparisons
- Benchmark reference lines
- Utilization analysis
- Demographic analysis

- Treatment analysis
- Admission and discharge analysis

The Executive Summary was designed to present the most important findings to a non-technical stakeholder.

Analytical Questions

The analysis was structured around the following questions:

1. **What is the overall 30-day hospital readmission rate?**
2. **How does the readmission rate differ between encounter-level and patient-level analysis?**
3. **Does previous inpatient utilization correspond with higher 30-day readmission?**
4. **Does previous emergency utilization correspond with higher 30-day readmission?**
5. **How does readmission vary across age groups?**
6. **Is length of hospital stay associated with readmission?**
7. **Does medication burden vary with readmission rate?**
8. **Does diagnosis burden vary with readmission rate?**
9. **How do insulin treatment and diabetes medication changes relate to observed readmission rates?**
10. **How do demographic and admission characteristics vary across readmission outcomes?**
11. **What proportion of patients experience repeated 30-day readmissions?**
12. **Can the available encounter characteristics support an exploratory predictive model?**

These questions provided the analytical framework for the Python, SQL, and Power BI stages of the project.

Data Dictionary

The following fields were used as key analytical variables in the project.

Field	Description	Analytical Use
encounter_id	Unique identifier for a hospital encounter	Encounter-level analysis
patient_nbr	Patient identifier	Patient-level analysis
age	Patient age group	Demographic segmentation
gender	Patient gender	Demographic analysis
race	Recorded race category	Demographic analysis
admission_type	Type of hospital admission	Admission analysis
admission_source	Source from which the patient was admitted	Admission pathway analysis
discharge_disposition	Patient disposition after discharge	Discharge analysis
time_in_hospital	Number of days spent in hospital	Length-of-stay analysis
num_lab_procedures	Number of laboratory procedures during encounter	Procedure-burden analysis
num_procedures	Number of non-laboratory procedures	Procedure analysis

Field	Description	Analytical Use
num_medications	Number of medications recorded	Medication-burden analysis
number_outpatient	Number of prior outpatient visits	Prior-utilization analysis
number_emergency	Number of prior emergency visits	Prior-utilization analysis
number_inpatient	Number of prior inpatient visits	Prior-utilization analysis
number_diagnoses	Number of diagnoses recorded	Diagnosis-burden analysis
insulin	Recorded insulin treatment status	Treatment analysis
change	Whether diabetes medication was changed	Treatment analysis
diabetesMed	Whether diabetes medication was prescribed	Treatment analysis
readmitted	Original readmission category	Source outcome
readmitted_30d	Binary indicator for readmission within 30 days	Primary analytical outcome
prior_inpatient_group	Grouped prior inpatient utilization	Readmission segmentation
prior_emergency_group	Grouped prior emergency utilization	Readmission segmentation
prior_outpatient_group	Grouped prior outpatient utilization	Readmission segmentation
length_of_stay_group	Grouped hospital length of stay	Readmission segmentation
medication_group	Grouped medication burden	Readmission segmentation
diagnosis_group	Grouped diagnosis burden	Readmission segmentation
lab_procedure_group	Grouped laboratory procedure burden	Readmission segmentation

6. Readmission Overview

The dataset contains three readmission categories:

Readmission Status	Encounters	Percentage
No readmission	54,864	53.91%
Readmitted >30 days	35,545	34.93%
Readmitted <30 days	11,357	11.16%

The primary analytical outcome is **readmission within 30 days**.

The overall encounter-level benchmark is therefore:

11.16%

This benchmark is used throughout the Power BI dashboard to provide context for subgroup comparisons.

7. Patient-Level Readmission

Because one patient can have multiple hospital encounters, encounter-level and patient-level readmission measures provide different perspectives.

Encounter-Level View

- Total encounters: **101,766**
- 30-day readmission encounters: **11,357**
- Readmission rate: **11.16%**

Patient-Level View

- Unique patients: **71,518**
- Patients with at least one 30-day readmission: **8,834**
- Patient-level readmission rate: **12.35%**

Among the 8,834 patients who experienced at least one 30-day readmission:

Number of 30-Day Readmissions	Patients
1	7,295
2	1,051
3	290
4+	198

A total of **1,539 patients** experienced two or more 30-day readmissions.

This represents **17.41% of patients who experienced at least one 30-day readmission**.

Encounter-Level vs Patient-Level Readmission

Because a patient can have multiple hospital encounters, encounter-level and patient-level readmission rates answer different questions.

Metric	Result
Total encounters	101,766
Unique patients	71,518
30-day readmission encounters	11,357
Encounter-level readmission rate	11.16%
Patients with ≥1 30-day readmission	8,834
Patient-level readmission rate	12.35%
Patients with 2+ 30-day readmissions	1,539

The **11.16% encounter-level rate** measures the proportion of hospital encounters classified as having a readmission within 30 days. The **12.35% patient-level rate** measures the proportion of unique patients who experienced at least one 30-day readmission.

These metrics should not be interpreted as interchangeable because they use different units of analysis. The patient-level metric also highlights the importance of repeated encounters and repeated readmissions within the dataset.

8. Prior Inpatient Utilization

Prior inpatient utilization showed a **substantial observed difference in 30-day readmission rates across utilization groups**.

Prior Inpatient Visits	Encounters	30-Day Readmission Rate
0	67,630	8.44%
1	19,521	12.92%
2	7,566	17.43%
3	3,411	20.29%
4+	3,638	30.70%

The readmission rate increased progressively across the prior-inpatient-utilization groups.

The **4+ group recorded a 30.70% readmission rate**, approximately 19.54 percentage points above the overall benchmark.

This pattern suggests that patients with a history of frequent inpatient utilization represent an important high-utilization subgroup for further investigation.

Important Interpretation

Prior inpatient utilization should be treated as an **associated characteristic**, not as a demonstrated cause of readmission.

Prior Inpatient Utilization — Key Comparison

Prior Inpatient Utilization	Readmission Rate
0 visits	8.44%
1 visit	12.92%
2 visits	17.43%
3 visits	20.29%
4+ visits	30.70%

Overall benchmark: 11.16%

*The readmission rate increases progressively across the prior-inpatient-utilization groups. The 4+ group is approximately **19.54 percentage points above the overall benchmark**.*

9. Prior Emergency Utilization

Prior emergency utilization showed a **substantial observed difference in 30-day readmission rates across utilization groups**.

Prior Emergency Visits	30-Day Readmission Rate
0	10.47%
1	14.35%
2	18.27%
3+	24.94%

The readmission rate increased from **10.47%** among encounters with no prior emergency visits to **24.94%** among encounters with 3+ prior emergency visits.

This represents a difference of approximately **14.47 percentage points**.

The finding indicates that previous emergency utilization can serve as an important descriptive marker of higher readmission burden in this dataset.

Prior Emergency Utilization — Key Comparison

Prior Emergency Utilization	Readmission Rate
0 visits	10.47%
1 visit	14.35%
2 visits	18.27%
3+ visits	24.94%

Overall benchmark: 11.16%

*Readmission rates increase across the prior-emergency-utilization groups. Encounters with 3+ prior emergency visits had a readmission rate **13.78 percentage points above the overall benchmark**.*

10. Age and Readmission

Readmission rates varied across age groups.

Age Group	30-Day Readmission Rate
0–10	1.86%
10–20	5.79%
20–30	14.24%
30–40	11.23%
40–50	10.60%
50–60	9.67%
60–70	11.13%
70–80	11.77%
80–90	12.08%
90–100	11.10%

The **20–30 age group recorded the highest observed rate at 14.24%**, although age groups at the extremes contain substantially fewer encounters than the major adult age groups.

Among the larger older-adult groups, readmission was approximately 11–12%.

Age should therefore be interpreted together with utilization, clinical complexity, and other patient characteristics rather than as an isolated characteristic of readmission risk.

11. Length of Stay

Readmission rates varied with length of hospital stay.

Length of Stay	30-Day Readmission Rate
1–2 days	9.14%

Length of Stay	30-Day Readmission Rate
3–4 days	11.17%
5–6 days	12.27%
7–9 days	13.50%
10+ days	12.76%

The rate generally increased as length of stay increased up to the 7–9 day group.

The 10+ day group showed a slightly lower rate than the 7–9 day group.

This suggests that length of stay is associated with readmission burden, but the relationship is not perfectly linear.

12. Medication Burden

Medication burden was examined using groups based on the number of medications recorded during an encounter.

The observed pattern showed increasing readmission rates across most medication-burden groups, followed by a slight reduction in the highest group.

Medication Group	30-Day Readmission Rate
1–5	7.48%
6–10	9.45%
11–15	10.91%
16–20	12.17%
21–30	12.87%
31+	12.43%

The pattern is consistent with **higher medication burden being associated with higher observed readmission rates across most groups**, although medication count may also reflect broader clinical complexity.

However, medication count is a utilization/complexity measure and should not be interpreted as an independent causal factor.

13. Diagnosis Burden

The number of diagnoses recorded during an encounter showed **differences in observed 30-day readmission rates across diagnosis-burden groups**.

Diagnosis Group	30-Day Readmission Rate
1–4	7.71%
5–6	9.75%
7–8	11.30%
9	12.38%
10+	14.78%

The rate increased across most diagnosis-burden categories.

The 10+ diagnosis group recorded a rate of **14.78%**, but this group contained only **115 encounters**.

Therefore, the estimate should be treated cautiously because of the small sample size.

14. Laboratory Procedure Burden

Laboratory procedure volume showed a **more modest difference in observed 30-day readmission rates across groups** than prior inpatient or emergency utilization.

Laboratory Procedure Group	30-Day Readmission Rate
1–30	10.10%
31–44	11.18%
45–57	11.43%
58+	11.90%

The readmission rate increased modestly as laboratory procedure volume increased.

The magnitude of this association was considerably smaller than the differences observed for prior inpatient and emergency utilization.

15. Diabetes Medication and Insulin Treatment

Diabetes Medication

The dashboard examined whether diabetes medication was recorded for the encounter.

Diabetes Medication	Encounters	30-Day Readmission Rate
No	23,403	9.60%
Yes	78,363	11.63%

Encounters where diabetes medication was recorded had a higher observed 30-day readmission rate (**11.63%**) than encounters where it was not recorded (**9.60%**).

This difference is descriptive and may reflect differences in underlying clinical complexity or treatment patterns rather than an independent effect of diabetes medication use.

Medication Change

Readmission rates were also compared between encounters where medication was changed and encounters where it was not.

The observed rates were:

- Medication change: **11.82%**
- No medication change: **10.59%**

Insulin Treatment

Insulin treatment categories also showed differences in observed readmission rates.

Insulin Status	30-Day Readmission Rate
No	10.04%
Steady	11.13%

Insulin Status	30-Day Readmission Rate
Up	12.99%
Down	13.90%

These treatment-related variables may reflect underlying disease severity, treatment complexity, or clinical decision-making rather than independently causing readmission.

16. Demographic Characteristics

Gender

Observed readmission rates were:

- Female: **11.25%**
- Male: **11.06%**

The difference was relatively small.

Race

Observed rates included:

Race Category	30-Day Readmission Rate
Caucasian	11.29%
African American	11.22%
Hispanic	10.41%
Asian	10.14%
Other	9.63%
Missing	8.27%

These are unadjusted descriptive differences.

They should not be interpreted as evidence that race itself causes readmission. Differences may reflect socioeconomic, clinical, healthcare-access, documentation, hospital-level, or other factors that are not fully represented in the dataset.

17. Admission Characteristics

Readmission rates were also evaluated by admission type, admission source, and discharge disposition.

Admission-type rates were relatively close to the overall **11.16% encounter-level benchmark** in the categories examined.

Emergency admissions recorded a readmission rate of approximately **11.52%**, compared with approximately **10.39% for elective admissions**.

These variables provide useful context for understanding the patient journey but do not independently establish risk.

18. Utilization as a High-Level Descriptive Marker

A consistent theme across the analysis is the relationship between previous healthcare utilization and subsequent readmission.

The major utilization findings were:

Utilization Measure	Lower Group	Higher Group	Difference
Prior inpatient visits	8.44%	30.70%	+22.26 pp
Prior emergency visits	10.47%	24.94%	+14.47 pp

These differences are substantially larger than several demographic differences observed in the dataset.

This makes prior healthcare utilization an important variable for descriptive segmentation and further analytical investigation.

19. SQL Analysis

SQL was used to reproduce and validate key findings and to transform the analysis into dashboard-ready summaries.

Key SQL outputs included:

- Overall readmission benchmark.
- Readmission rates by prior inpatient utilization.
- Readmission rates by prior emergency utilization.
- Patient-level readmission statistics.
- Repeated readmission burden.
- Utilization segmentation.
- Readmission differences versus the overall benchmark.

SQL Validation

The SQL analysis reproduced the core benchmark:

11.16% encounter-level 30-day readmission rate

The patient-level analysis identified:

71,518 unique patients

and:

8,834 patients with at least one 30-day readmission.

SQL therefore served as a **cross-tool validation layer**, reproducing key findings from the Python analysis before they were presented in Power BI.

20. Power BI Dashboard

The Power BI dashboard was designed to communicate the findings to a non-technical healthcare or business audience.

The dashboard contains:

Executive Summary

The Executive Summary page includes:

- Overall 30-Day Readmission Rate
- Patient-Level Readmission Rate
- Total Encounters
- Unique Patients
- 30-Day Readmissions
- Patients With 30-Day Readmission
- Prior inpatient utilization analysis
- Prior emergency utilization analysis
- Key analytical insight

Detailed Analysis Pages

Additional pages examine:

- Age
- Length of stay
- Medication burden
- Diagnosis burden
- Insulin treatment
- Diabetes medication
- Medication change
- Race
- Gender
- Prior outpatient utilization
- Laboratory procedure burden
- Admission source
- Discharge disposition
- Admission type

A benchmark line of **11.16%** is used in the readmission-rate visuals as a **descriptive reference point** for comparing subgroup rates with the overall encounter-level readmission rate.

21. Key Findings

Finding 1 — Prior inpatient utilization shows a substantial observed difference

The 30-day readmission rate increased from 8.44% for encounters with no prior inpatient visits to 30.70% for encounters with 4+ prior inpatient visits.

Finding 2 — Prior emergency utilization also shows a substantial observed difference

Readmission increased from 10.47% among encounters with no prior emergency visits to 24.94% among encounters with 3+ prior emergency visits.

Finding 3 — Patient-level and encounter-level rates differ

The encounter-level readmission rate was **11.16%**, while the patient-level rate was **12.35%**.

This demonstrates why healthcare analytics should distinguish between encounter-level and patient-level measures.

Finding 4 — Repeat readmission is concentrated among a smaller patient group

Among patients with at least one 30-day readmission, **17.41% experienced two or more 30-day readmissions**.

This subgroup may warrant further analysis of utilization patterns and clinical complexity.

Finding 5 — Several complexity measures show higher observed rates at higher levels

Length of stay, medication burden, and diagnosis burden generally showed higher readmission rates at higher levels.

These variables may act as indicators of patient or encounter complexity.

Finding 6 — Demographic differences were comparatively smaller

Gender differences were relatively small, while race showed moderate unadjusted variation.

These findings should not be interpreted independently of clinical and socioeconomic context.

22. Analytical Model

A logistic regression model was developed as an exploratory predictive model.

The modeling workflow used a **patient-level data split** to reduce leakage caused by multiple encounters belonging to the same patient.

Model Performance

Metric	Result
ROC-AUC	0.6516
Average Precision	0.1824
Brier Score	0.2301
Precision at 0.50 threshold	0.1572
Recall at 0.50 threshold	0.5086
F1 Score	0.2401
Percentage flagged	34.54%

The model demonstrated **modest discrimination**.

Calibration analysis indicated that the predicted probabilities were not sufficiently reliable for direct clinical deployment.

Therefore, the model should be considered an **exploratory analytical exercise rather than a clinical prediction tool**.

23. Business Interpretation

The findings suggest several areas that healthcare organizations could investigate further.

1. High-utilization patient segmentation

Patients with repeated prior inpatient or emergency utilization could be segmented for additional analysis.

2. Transitional-care analysis

Higher observed readmission rates among patients with greater prior utilization may justify further investigation of discharge planning, follow-up coordination, medication reconciliation, and post-discharge support.

3. Repeat-readmission monitoring

Patients experiencing multiple readmissions represent a smaller but important subgroup for longitudinal analysis.

4. Complexity-aware reporting

Medication burden, diagnosis burden, and length of stay can provide additional context when examining readmission patterns.

5. Data-driven resource planning

Utilization patterns can potentially help healthcare organizations identify populations requiring deeper review.

These are **potential analytical applications**, not claims that the dataset proves these interventions would reduce readmissions.

Business Recommendations

Based on the observed associations in the dataset, the following areas could be considered for further operational investigation:

1. Prioritize high-utilization patients for review

Encounters involving patients with repeated prior inpatient or emergency utilization show substantially higher observed readmission rates. Healthcare organizations could investigate whether these patients would benefit from enhanced discharge planning, follow-up coordination, or care-management review.

2. Strengthen post-discharge follow-up processes

The elevated observed readmission rates among higher-utilization groups indicate that continuity of care after discharge is an area that could be investigated further, particularly for patients with complex utilization histories.

3. Monitor patients with high clinical burden

Higher medication and diagnosis burden groups generally showed higher observed readmission rates. These characteristics could be explored as part of broader patient-complexity assessments rather than being treated as independent causal factors.

4. Use dashboards for ongoing monitoring

The Power BI dashboard demonstrates how readmission metrics can be monitored across utilization, demographic, treatment, admission, and discharge characteristics. Similar dashboards could support recurring operational reporting.

5. Validate findings with current and local data

Because this analysis uses a historical U.S. dataset, the observed relationships should not be directly applied to current healthcare settings or Indian patient populations without validation using contemporary local data.

***Important:** These recommendations are based on observational associations in the dataset. They should be treated as areas for further investigation rather than evidence that any particular intervention will reduce readmissions.*

24. Limitations

Several limitations should be considered when interpreting this project.

Historical Dataset

The data represents U.S. hospital encounters from approximately **1999–2008**.

Healthcare delivery, treatment protocols, technology, coding practices, and patient populations have changed substantially since then.

Geographic Limitation

The dataset represents U.S. healthcare encounters and should not be assumed to represent current Indian hospitals or patients.

Observational Data

The analysis identifies associations.

It does not establish that a variable causes readmission.

Encounter-Level Structure

Patients can contribute multiple encounters.